

DD: OSR-RMA-01-FLOW-EQU — Equipment Upgrade

Developer implementation guide: DD_OSR-RMA-01-FLOW-EQU-DEV_IMPL_GUIDE-v1.0.md.

1. Purpose

This rewritten DD is the developer-facing version of the operator BPMN flow DD_OSR-RMA-01-FLOW-EQU-v1.0.md.

Verdict: ■ New | **Wave:** 2 | **Group:** OSR (Operator Stock Registry) | **Version:** v1.0 | **Module:** Equipment Upgrade Flow
Satellite of: OSR-RMA-01 Equipment Swap & RMA Per-flow satellite for Equipment Upgrade (swap_kind=EQU_UPGRADE). Owns the BPMN EquipmentUpgrade_EQU_WIK.bpmn + per-flow REST endpoint + per-flow Kafka payload shape + flow-specific Drools rules + worked end-to-end example. Reuses framework workers, messages, user tasks, history schema, and FOUNDATION_AUTH groups. v1.0 scope: KE WIK only. Always chargeable; always pay-first.

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/08_osr_equipment/DD_OSR-RMA-01-FLOW-EQU-v1.0.md
Version	v1.0
DD type	operator BPMN flow
BPMN/process keys	Not explicitly defined
Drools packages	Not explicitly defined

4. Runtime Contracts To Implement

4.1 APIs

- GET /api/swap-requests/swap_01HZ_MUTHONI_UPG
- POST /api/billable-events
- POST /api/equipment-instances/inst_01HZ_CM_MUTHONI_DOCSIS30/lifecycle-events
- POST /api/equipment-instances/inst_01HZ_CM_MUTHONI_DOCSIS31/lifecycle-events
- POST /api/swap-requests/equ
- POST /broadhub-adapter/v1/serial/rebind
- POST /broadhub-adapter/v1/serial/unbind

4.2 Tables

- No CREATE TABLE block is explicitly declared in this DD.

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;
- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- bil01-charge-pre-authorize
- contractor-slot-reserve
- nrb-cmts-01
- payment-confirmed

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- No domain events are explicitly declared in this DD.

Event guidance:

- Write events through the transactional outbox.
- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.
4. Implement idempotency and in-flight operation checks where the DD requires them.
5. Implement Drools fact mapping and package deployment where rules are used.
6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.
7. Implement worker-topic handlers using owning module APIs.
8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.
10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.

Concern	Recommendation
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

Verdict: ■ New | **Wave:** 2 | **Group:** OSR (Operator Stock Registry) | **Version:** v1.0 | **Module:** Equipment Upgrade Flow

Satellite of: OSR-RMA-01 Equipment Swap & RMA

Per-flow satellite for Equipment Upgrade (swap_kind=EQU_UPGRADE). Owns the BPMN EquipmentUpgrade_EQU_WIK.bpmn + per-flow REST endpoint + per-flow Kafka payload shape + flow-specific Drools rules + worked end-to-end example. Reuses framework workers, messages, user tasks, history schema, and FOUNDATION_AUTH groups.

v1.0 scope: KE WIK only. Always chargeable; always pay-first.

1. What this flow is

A customer requests a tech upgrade (DOCSIS 3.0 cable modem → DOCSIS 3.1, Huawei ONT v1 → v2, SD STB → 4K STB), OR the operator initiates a mandatory replacement campaign (e.g., end-of-life of an SKU forces fleet-wide replacement). Either way, a contractor visits, swaps the old device for the new upgraded one, and the old device returns to main warehouse for inspection (NOT defective — typically re-deployable after testing).

The defining property of EQU compared to defect swaps (SWAP-HFC, SWAP-GPON): the source instance is **not** defective. After recovery, it goes through main-warehouse inspection; if good, it re-enters the available pool (IN_MAIN_WAREHOUSE). For SWAP-HFC/GPON, the source is defective and typically heads for RMA-to-vendor.

EQU is **always** chargeable: there's an upgrade fee defined in the BIL-01 charge catalog, plus a possible deposit delta (if the upgraded SKU carries a higher equipment deposit, the customer pays the difference; lower deposit = customer gets a partial refund queued for post-completion). Per workshop 2026-04-28: "Customer emails Call Center → CC gives advice (price + package) → customer makes payment → BO Dispatcher operates the change in Broadhub → WO scheduled." The pay-first sequence is baked in.

When the upgrade involves GPON, the HomePass network_path terminal node serial is updated at TARGET_PLACED (same as SWAP-GPON). For HFC EQU, no HomePass update (HFC equipment isn't in HomePass.network_path).

What this flow does NOT do

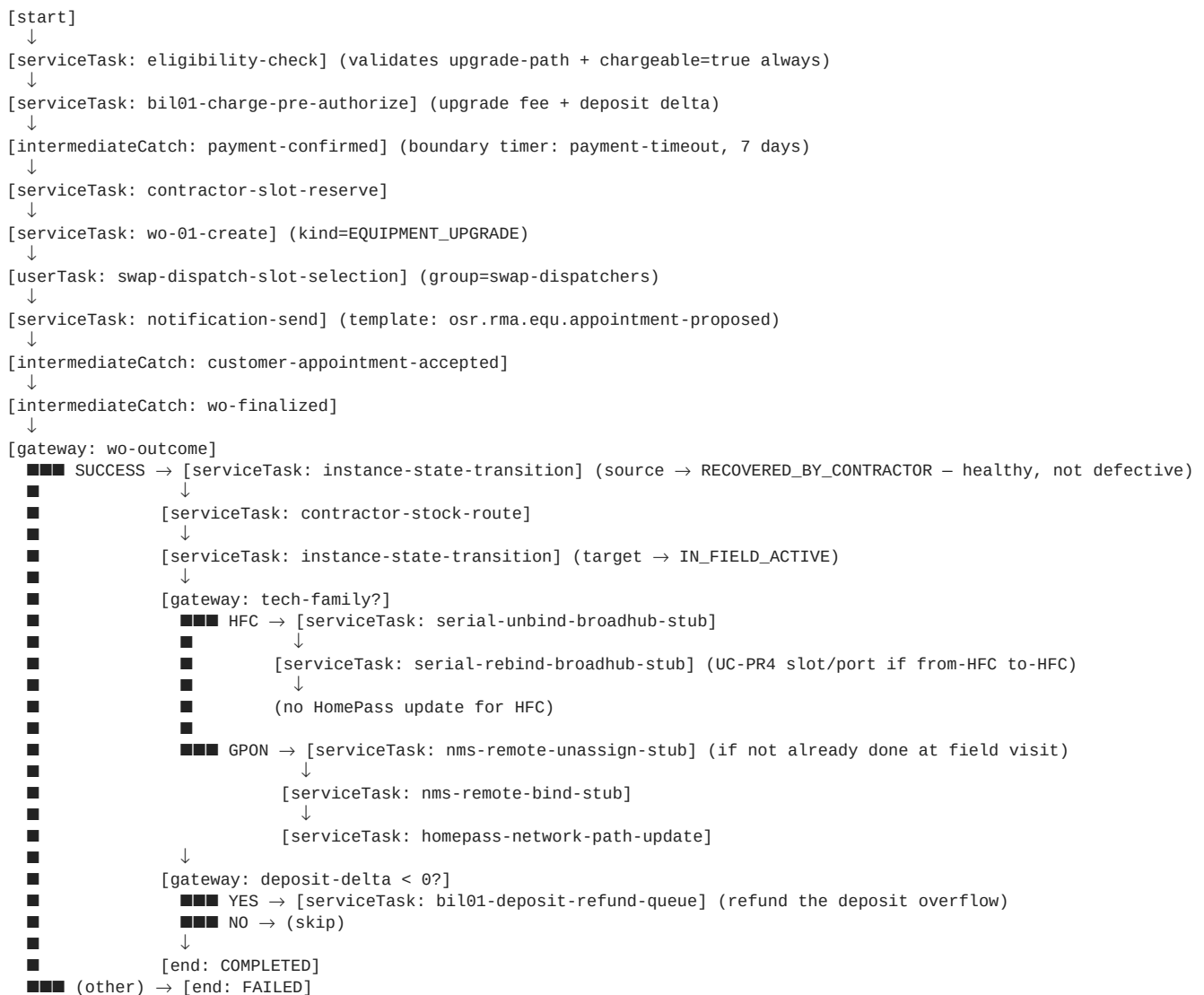
- **Does not change the subscription package or service class.** That's SUB-WF-UPGRADE-01 (subscription upgrade workflow). EQU is the equipment-side companion: when a subscription upgrade requires new hardware, SUB-WF-UPGRADE-01 creates an EQU swap-request. EQU then runs to completion; SUB-WF-UPGRADE-01 catches SwapRequestCompleted and proceeds with its own steps (price change, service activation, etc.). For pure equipment upgrades that don't change subscription (rare — e.g., voluntary modem refresh), EQU runs standalone.
- **Does not handle defective-equipment swaps.** Use SWAP-HFC or SWAP-GPON.
- **Does not handle termination-time equipment retrieval.** Use EQP.
- **Does not transition the source instance to IN_FIELD_DEFECTIVE.** The source is healthy; on recovery, it goes to RECOVERED_BY_CONTRACTOR then IN_CONTRACTOR_STOCK (van) then RETURNED_TO_MAIN_WAREHOUSE for inspection.

Glossary additions (specific to this flow)

Term	Meaning
EQU (Equipment Upgrade)	Job Type code in wo_lifecycle.md. WO is wo_kind=EQUIPMENT_UPGRADE.
Upgrade fee	Customer-paid fee for the upgrade transaction, defined per upgrade-path in BIL-01 charge catalog. Typical KES range 1,500–5,000 depending on the upgrade.
Deposit delta	Difference between new SKU deposit and old SKU deposit. Positive = customer pays extra; negative = customer gets refund queued for post-completion. Captured in flow_payload.depositDelta.
Upgrade path	A (upgrade_from_sku, upgrade_to_sku) pair declared as valid in PLM-CFG-06 operator config. Not all upgrades are valid (e.g., DOCSIS 3.1 cable modem → GPON ONT is a tech change, not an upgrade — that's a subscription relocation). v1.0 expects the caller to pass valid SKUs; eligibility worker validates against PLM-CFG-06.
Re-deployable	Status of a recovered EQU source: physically intact, functionally tested OK at main warehouse, ready to be re-dispatched to another customer install or another EQU.

2. BPMN — EquipmentUpgrade_EQU_WIK.bpmn

2.1 Process shape



2.2 BPMN XML — key snippets

Pay-first is mandatory; no chargeable-gateway branch:

```

<bpmn:serviceTask id="Task_PreAuthCharge" name="Pre-authorize upgrade fee + deposit delta"
    camunda:type="external" camunda:topic="bil01-charge-pre-authorize">
  <bpmn:extensionElements>
    <camunda:inputOutput>
      <camunda:inputParameter name="swapRequestId">${swapRequestId}</camunda:inputParameter>
      <camunda:inputParameter name="accountId">${accountId}</camunda:inputParameter>
      <camunda:inputParameter name="chargeCode">EQUIPMENT_UPGRADE_FEE</camunda:inputParameter>
      <camunda:inputParameter name="customerChargeAmount">${flowPayload.upgradeFee + (flowPayload.depositDelta > 0 ? flowPayload.depositDelta : 0)}</camunda:inputParameter>
      <camunda:inputParameter name="flowPayload">${flowPayload}</camunda:inputParameter>
    </camunda:inputOutput>
  </bpmn:extensionElements>
</bpmn:serviceTask>

```

Tech-family decision gateway after target placement:

```

<bpmn:exclusiveGateway id="Gateway_TechFamily" name="Tech family of target" />
<bpmn:sequenceFlow id="Flow_HfcUpgrade" sourceRef="Gateway_TechFamily" targetRef="Task_UnbindBroadhub">
  <bpmn:conditionExpression>${flowPayload.targetTechFamily == 'HFC'}</bpmn:conditionExpression>
</bpmn:sequenceFlow>
<bpmn:sequenceFlow id="Flow_GponUpgrade" sourceRef="Gateway_TechFamily" targetRef="Task_NmsUnassign">
  <bpmn:conditionExpression>${flowPayload.targetTechFamily == 'GPON'}</bpmn:conditionExpression>
</bpmn:sequenceFlow>

```

2.3 Per-operator BPMN composition

KE WIK ships EquipmentUpgrade_EQU_WIK.bpmn. WUG/WTZ/YASSN variants TBD pending operator-specific upgrade catalog workshops.

3. Flow_payload shape (EQU_UPGRADE)

```

{
  "upgradeFromSku": "WIK-CM-CISCO-DPC3848",
  "upgradeToSku": "WIK-CM-CISCO-CBR8",
  "upgradeFee": 3500.00,
  "depositDelta": 500.00,
  "oldDepositAmount": 2000.00,
  "newDepositAmount": 2500.00,
  "targetTechFamily": "HFC",
  "sourceTechFamily": "HFC",
  "isCrossTechUpgrade": false,
  "upgradePathRef": "UPGRADE_PATH_DOCSIS30_TO_DOCSIS31",
  "initiatedBy": "CUSTOMER_REQUESTED",
  "subscriptionUpgradeRef": "subupg_01HZ_MUTHONI_TIER_BUMP"
}

```

For HFC swaps, the flow_payload also carries the HFC slot/port preservation fields (same as FLOW-SWAP-HFC §3 if targetTechFamily=HFC). For GPON, the OLT F/S/P fields (same as FLOW-SWAP-GPON §3 if targetTechFamily=GPON). The eligibility worker merges these in from the source instance's network_slot_metadata.

Field meanings:

Field	Sourced from	Purpose
upgradeFromSku, upgradeToSku	Caller (in request body)	Identify the upgrade path; validated against PLM-CFG-06
upgradeFee	BIL-01 charge catalog lookup by upgradePathRef	Customer-facing fee
depositDelta	newDepositAmount - oldDepositAmount	Positive = customer pays extra; negative = refund queued post-completion
targetTechFamily, sourceTechFamily	PLM-CFG-06 equipment_kind classification	Drives BPMN tech-family gateway
isCrossTechUpgrade	Boolean derived from family comparison	TRUE = HFC↔GPON, generally rejected (out of v1.0); FALSE = same-family upgrade
upgradePathRef	PLM-CFG-06 operator config	The named upgrade path; resolves fee + DB-stored eligibility
initiatedBy	Caller	CUSTOMER_REQUESTED / OPERATOR_CAMPAIGN / SUB_WF_UPGRADE_DRIVEN
subscriptionUpgradeRef	When created by SUB-WF-UPGRADE-01	Links back to the subscription upgrade workflow

4. Rules (EQU-specific)

Rule	Description
R-EQU-1	upgradeFromSku MUST equal source instance's sku_id. Eligibility worker enforces.
R-EQU-2	(upgradeFromSku, upgradeToSku) MUST be declared as a valid upgrade path in PLM-CFG-06 operator config. Eligibility worker rejects with failureCode=INVALID_UPGRADE_PATH otherwise. v1.0 KE WIK paths: DOCSIS 3.0 → DOCSIS 3.1 cable modem, ONT v1 → ONT v2, SD STB → 4K STB. Cross-tech (HFC→GPON or vice versa) NOT supported as upgrade in v1.0 (use subscription relocation instead).
R-EQU-3	chargeable=TRUE always. Pay-first gate is mandatory. Free EQU (operator-paid mandatory upgrade campaigns where the customer doesn't pay) handled out of v1.0 scope — would require a chargeable=FALSE override + a OPERATOR_CAMPAIN_FREE reason code.
R-EQU-4	Source instance MUST be IN_FIELD_ACTIVE. EQU does not handle defective-source (that's SWAP-* with reason CUSTOMER_REPORTED_DEFECT).
R-EQU-5	After recovery, source instance transitions through RECOVERED_BY_CONTRACTOR → IN_CONTRACTOR_STOCK → RETURNED_TO_MAIN_WAREHOUSE (per main-warehouse re-inspection path). NOT IN_FIELD_DEFECTIVE. The transition to IN_MAIN_WAREHOUSE happens after main-warehouse inspection (out of EQU scope; main-warehouse ops handles).
R-EQU-6	When depositDelta > 0: included in the pay-first billable event amount (customer pays upgrade fee + deposit delta in one transaction). When depositDelta < 0: refund queued POST-completion via a separate BIL-01 billable event with chargeCode=DEPOSIT_REFUND, amount=abs(depositDelta), direction=REFUND_TO_CUSTOMER. R-OSR-RMA-11c (framework) governs.
R-EQU-7	For HFC upgrade where both source and target are HFC and on the same CMTS: UC-PR4 slot/port preservation applies (same as SWAP-HFC R-HFC-5). For cross-CMTS HFC upgrade (rare — usually only when the customer's premises migrated to a different CMTS chassis), preservation does not apply and the rebind uses the target's new slot/port.
R-EQU-8	For GPON upgrade: HomePass network_path terminal-node update is mandatory (same as SWAP-GPON R-GPON-5). The new ONT serial replaces the old serial in the HomePass terminal node.
R-EQU-9	If caller is SUB-WF-UPGRADE-01 (subscription-driven upgrade), flow_payload.subscriptionUpgradeRef MUST be populated. SUB-WF-UPGRADE-01 waits for SwapRequestCompleted before proceeding with its own next step. EQU failure cascades to SUB-WF-UPGRADE-01 failure (subscription rolls back to previous tier; the equipment stays as-is).
R-EQU-10	The target instance after a successful EQU is IN_FIELD_ACTIVE bound to the customer's account, same as a fresh install. The customer's subscription package (if upgraded by SUB-WF-UPGRADE-01) catches up via separate provisioning steps; this flow is equipment-only.

5. REST API

POST /api/swap-requests/equ HTTP/1.1
Authorization: Bearer <cc-agent-jwt>
Content-Type: application/json

```
{
  "operatorCode":      "WIK",
  "accountId":         "acct_01HZ_MUTHONI",
  "subscriptionId":    "sub_01HZ_MUTHONI_TRIPLEPLAY",
  "sourceInstanceId":  "inst_01HZ_CM_MUTHONI_DOCSIS30",
  "targetInstanceId":  "inst_01HZ_CM_MUTHONI_DOCSIS31",
  "reasonCode":        "CUSTOMER_REQUESTED_TECH_CHANGE",
  "initiatedByUserId": "cc-agent-wik-012",
  "flowPayload": {
    "upgradeFromSku": "WIK-CM-CISCO-DPC3848",
    "upgradeToSku":   "WIK-CM-CISCO-CBR8",
    "upgradePathRef": "UPGRADE_PATH_DOCSIS30_TO_DOCSIS31",
    "initiatedBy":    "CUSTOMER_REQUESTED",
    "subscriptionUpgradeRef": null
  },
  "notes": "Customer requested upgrade to gigabit tier; CBR8 modem ready in Franchise A stock"
}
```

→ 201 Created

```
{
  "swapRequestId":      "swap_01HZ_MUTHONI_UPG",
```

```

"operatorCode": "WIK",
"swapKind": "EQU_UPGRADE",
"currentState": "ELIGIBILITY_CHECK",
"accountId": "acct_01HZ_MUTHONI",
"subscriptionId": "sub_01HZ_MUTHONI_TRIPLEPLAY",
"sourceInstanceId": "inst_01HZ_CM_MUTHONI_DOCSIS30",
"targetInstanceId": "inst_01HZ_CM_MUTHONI_DOCSIS31",
"reasonCode": "CUSTOMER_REQUESTED_TECH_CHANGE",
"initiatedByUserId": "cc-agent-wik-012",
"initiatedAt": "2026-07-05T11:00:00Z",
"bpmnProcessInstanceId": "camunda-pi-ad6e3f1a"
}

```

After eligibility resolves the upgrade fee + deposit delta:

```
GET /api/swap-requests/swap_01HZ_MUTHONI_UPG
```

→ 200 OK

```

{
  "swapRequestId": "swap_01HZ_MUTHONI_UPG",
  "currentState": "AWAITING_PAYMENT",
  "chargeable": true,
  "chargeCode": "EQUIPMENT_UPGRADE_FEE",
  "customerChargeAmount": 4000.00,
  "billableEventId": "bevt_01HZ_MUTHONI_UPG_CHARGE",
  "flowPayload": {
    "upgradeFromSku": "WIK-CM-CISCO-DPC3848",
    "upgradeToSku": "WIK-CM-CISCO-CBR8",
    "upgradeFee": 3500.00,
    "depositDelta": 500.00,
    "oldDepositAmount": 2000.00,
    "newDepositAmount": 2500.00,
    "targetTechFamily": "HFC",
    "sourceTechFamily": "HFC",
    "isCrossTechUpgrade": false,
    "upgradePathRef": "UPGRADE_PATH_DOCSIS30_TO_DOCSIS31",
    "initiatedBy": "CUSTOMER_REQUESTED",
    "cmts": "nrb-cmts-01", "slot": 5, "port": 8, "authzBound": true, "doubleConfig": false,
    "sourceCmMac": "00:11:22:AA:BB:CC", "targetCmMac": "00:11:22:DD:EE:FF",
    "isHybridChassis": false, "isStaticIp": false
  }
}

```

(customerChargeAmount = upgradeFee + depositDelta = 3500 + 500 = 4000.)

6. Kafka events emitted

6.1 SwapRequestAwaitingPayment

```

{
  "eventType": "SwapRequestAwaitingPayment",
  "aggregateId": "swap_01HZ_MUTHONI_UPG",
  "timestamp": "2026-07-05T11:00:42.000Z",
  "payload": {
    "operatorCode": "WIK",
    "swapRequestId": "swap_01HZ_MUTHONI_UPG",
    "swapKind": "EQU_UPGRADE",
    "accountId": "acct_01HZ_MUTHONI",
    "chargeCode": "EQUIPMENT_UPGRADE_FEE",
    "customerChargeAmount": 4000.00,
    "upgradeFee": 3500.00,
    "depositDelta": 500.00,
    "billableEventId": "bevt_01HZ_MUTHONI_UPG_CHARGE",
    "paymentDeadline": "2026-07-12T11:00:42.000Z"
  }
}

```

6.2 SwapRequestCompleted

```

{
  "eventType": "SwapRequestCompleted",
  "aggregateId": "swap_01HZ_MUTHONI_UPG",
  "timestamp": "2026-07-10T16:30:00.000Z",
  "payload": {
    "operatorCode": "WIK",
    "swapRequestId": "swap_01HZ_MUTHONI_UPG",
    "swapKind": "EQU_UPGRADE",
    "accountId": "acct_01HZ_MUTHONI",
    "subscriptionId": "sub_01HZ_MUTHONI_TRIPLEPLAY",
  }
}

```



```

    "sourceInstanceId": "inst_01HZ_CM_MUTHONI_DOCSIS30",
    "targetInstanceId": "inst_01HZ_CM_MUTHONI_DOCSIS31",
    "recoveryContractorId": "contractor_ke_nairobi_a",
    "woId": "wo_01HZ_MUTHONI_UPG",
    "chargeCode": "EQUIPMENT_UPGRADE_FEE",
    "customerChargeAmount": 4000.00,
    "upgradeFee": 3500.00,
    "depositDelta": 500.00,
    "paidAt": "2026-07-05T14:30:00Z",
    "upgradePathRef": "UPGRADE_PATH_DOCSIS30_TO_DOCSIS31",
    "closedAt": "2026-07-10T16:30:00+03:00"
  }
}

```

7. Worked end-to-end example — Muthoni's DOCSIS 3.0 → 3.1 upgrade

Customer: Muthoni, WIK HFC triple-play subscriber, on a 100 Mbps DOCSIS 3.0 plan. Wants to upgrade to the new 500 Mbps gigabit plan, which requires a DOCSIS 3.1 cable modem. Account acct_01HZ_MUTHONI. Old modem inst_01HZ_CM_MUTHONI_DOCSIS30 (Cisco DPC3848, CM-MAC 00:11:22:AA:BB:CC), on Nairobi CMTS nrb-cmts-01, slot 5, port 8. New modem inst_01HZ_CM_MUTHONI_DOCSIS31 (Cisco CBR8, CM-MAC 00:11:22:DD:EE:FF) is in Franchise A's van stock. PLM-CFG-06 declares upgrade path UPGRADE_PATH_DOCSIS30_TO_DOCSIS31: fee KES 3,500, new deposit KES 2,500 (old KES 2,000), delta +500. Total customer charge: 4,000.

Day 0 (2026-07-05) — 11:00: Muthoni calls CC. CC agent confirms package availability + upgrade fee. Muthoni agrees. CC posts:

```

POST /api/swap-requests/equ
{"operatorCode":"WIK","accountId":"acct_01HZ_MUTHONI","subscriptionId":"sub_01HZ_MUTHONI_TRIPLEPLAY",
 "sourceInstanceId":"inst_01HZ_CM_MUTHONI_DOCSIS30","targetInstanceId":"inst_01HZ_CM_MUTHONI_DOCSIS31",
 "reasonCode":"CUSTOMER_REQUESTED_TECH_CHANGE","initiatedByUserId":"cc-agent-wik-012",
 "flowPayload":{"upgradeFromSku":"WIK-CM-CISCO-DPC3848","upgradeToSku":"WIK-CM-CISCO-CBR8",
  "upgradePathRef":"UPGRADE_PATH_DOCSIS30_TO_DOCSIS31","initiatedBy":"CUSTOMER_REQUESTED"}}
→ 201 swap_01HZ_MUTHONI_UPG

```

BPMN starts. Eligibility worker validates path against PLM-CFG-06 (DOCSIS 3.0 → 3.1, declared, fee = 3500, deposit delta = +500), reads source instance metadata (slot/port present), builds full flow_payload with HFC fields, sets chargeable=true, chargeCode=EQUIPMENT_UPGRADE_FEE, customerChargeAmount=4000.00. State → AWAITING_PAYMENT.

Day 0 — 11:00:42: Worker bil01-charge-pre-authorize fires:

```

POST /api/billable-events
{"accountId":"acct_01HZ_MUTHONI","chargeCode":"EQUIPMENT_UPGRADE_FEE","amount":4000.00,
 "direction":"CHARGE_TO_CUSTOMER","contextRef":{"kind":"swap_request","id":"swap_01HZ_MUTHONI_UPG"},
 "breakdownDetail":{"upgradeFee":3500.00,"depositDelta":500.00}}
→ 201 {"billableEventId":"bevt_01HZ_MUTHONI_UPG_CHARGE","status":"AWAITING_PAYMENT"}

```

Kafka emits SwapRequestAwaitingPayment. NOT-01 sends Muthoni: *Hi Muthoni, your gigabit upgrade is ready. Pay KES 4,000 via M-Pesa to 123456, reference UPG-MUTHONI. Once paid, we'll schedule the tech visit.*

Day 0 — 14:30: Muthoni pays KES 4,000 via M-Pesa. BIL-01 confirms. BillableEventPaid Kafka event fires. BPMN message catch payment-confirmed. State AWAITING_PAYMENT → AWAITING_SLOT.

Day 0 — 14:31: Worker contractor-slot-reserve → EM-02 → slot reserved with contractor_ke_nairobi_a, scheduled 2026-07-10 14:00.

Day 0 — 14:32: Worker calls WO-01 → wo_01HZ_MUTHONI_UPG created with wo_kind=EQUIPMENT_UPGRADE. NOT-01 → Muthoni confirms slot.

Day 5 (2026-07-10) — 14:05: Tech arrives with new CBR8. Confirms WO. Removes old modem; installs new. Submits user task (no defect needed — EQU doesn't require defect confirmation, only "swap done"). Workers fire:

```

POST /api/equipment-instances/inst_01HZ_CM_MUTHONI_DOCSIS30/lifecycle-events
{"toState":"RECOVERED_BY_CONTRACTOR","toLocationId":"WIK-VAN-contractor_ke_nairobi_a",
 "reasonCode":"CONTRACTOR_RECOVERED_FROM_FIELD","contractorId":"contractor_ke_nairobi_a",
 "originRefKind":"swap_request","originRefId":"swap_01HZ_MUTHONI_UPG",
 "actorUserId":"tech_franchise_a_03","expectedEventSequence":4}
→ 201

```

Old modem in Franchise A's van. Target placed:

```

POST /api/equipment-instances/inst_01HZ_CM_MUTHONI_DOCSIS31/lifecycle-events
{"toState":"IN_FIELD_ACTIVE","toAccountId":"acct_01HZ_MUTHONI",
 "reasonCode":"WO_INSTALL_FINALIZED","contractorId":"contractor_ke_nairobi_a",

```



```
"originRefKind":"swap_request", "originRefId":"swap_01HZ_MUTHONI_UPG",  
"actorUserId":"tech_franchise_a_03", "expectedEventSequence":3}  
→ 201
```

Tech-family gateway: HFC. Broadhub unbind old + rebind new at SAME slot/port (nrb-cmts-01, slot 5, port 8):

```
POST /broadhub-adapter/v1/serial/unbind {"cmMac":"00:11:22:AA:BB:CC", "slotPort":{"cmts":"nrb-cmts-01", "slot":5, "port":8}}  
→ 200 {"unbindOk":true}
```

```
POST /broadhub-adapter/v1/serial/rebind {"cmMac":"00:11:22:DD:EE:FF", "slotPort":{"cmts":"nrb-cmts-01", "slot":5, "port":8}, "  
→ 200 {"rebindOk":true}
```

Deposit-delta gateway: +500 (positive), already collected at pay-first; no refund queued. State → COMPLETED.

Day 5 — 16:30: Kafka emits SwapRequestCompleted. NOT-01 sends Muthoni: *Hi Muthoni, your gigabit upgrade is complete. Enjoy your new 500 Mbps speeds. Speed test: <https://wik.example/speedtest>*

Day 6+: Old DOCSIS 3.0 modem on Franchise A's van. At next Franchise A inventory cycle, the modem is shipped to main warehouse with status RETURNED_TO_MAIN_WAREHOUSE. Main-warehouse tech tests it, confirms it's functionally good, transitions it to IN_MAIN_WAREHOUSE — available for re-deployment to a future DOCSIS 3.0 install or another EQU's reverse-upgrade path (rare). All transitions handled by main-warehouse ops via OSR-INSTANCE-01 APIs, not by this flow.

8. Workshop sources

- **Workshop 2026-04-28 (Customer Management deep dive — Customer Account Management screen)** — Upgrade/downgrade operator flow: Customer emails Call Center → CC gives advice (price + package) → customer makes payment → BO Dispatcher operates the change in Broadhub → WO scheduled. Drives R-EQU-3 (always chargeable, pay-first) and the §2 BPMN sequence.
- **Workshop 2026-04-28 (Billing deep dive)** — Wallet funding gate for contractor WOs: tech dispatched only after wallet topped up. R-EQU-3 codifies for EQU.
- **Workshop 2026-04-30 (Provisioning deep dive)** — UC-PR4 slot/port preservation: applies to HFC EQU when source and target are same-CMTS HFC. R-EQU-7 codifies.
- **wo_lifecycle.md Job Type catalog** — EQU (Equipment Upgrade) defined. WO wo_kind=EQUIPMENT_UPGRADE.
- **DD_HomePass-CFG §"Equipment / ONT binding"** — HomePass.network_path terminal node code update for GPON. R-EQU-8 codifies for GPON EQU.
- **DD_PLM-CFG-06 Equipment Type Catalog** — owns the equipment_sku table + (future, pending) operator-config upgrade path declarations. R-EQU-2 codifies.
- **BP-08 Equipment Swap (RMA) GAP analysis** — Equipment upgrade is one of the four target swap categories; this satellite addresses it.